# Improving Reasoning in Small Language Models via Structured Post-Training and Domain Specialization

Elias Mir | 341277 | elias.mir@epfl.ch
Valentin Planes | 409205 | valentin.planes@epfl.ch
Martin Daigne | 340936 | martin.daigne@epfl.ch
Tomas Garate Anderegg | 313984 | tomas.garateanderegg@epfl.ch
MVTE

## Abstract

Small language models struggle with structured reasoning despite strong general pretraining. We present a post-training pipeline applied to Qwen3-1.7B [1] targeting four reasoning domains: mathematical problem solving, general knowledge, multilingual understanding, and safety. Each domain is addressed by a dedicated individual model trained with methods suited to the verification structure of its benchmark, while a joint group model balances generalization across all four.

## 1 Introduction

Small language models still lag behind larger systems on structured reasoning, factual recall, multilingual transfer and safety-sensitive judgement. Reinforcement-based post-training methods such as DeepSeek-R1 [2] show that reasoning can be improved after pretraining, but no single strategy suits all domains: mathematics benefits from verifiable numeric rewards, knowledge tasks require broad factual coverage [3], multilingual evaluation exposes cultural distribution shifts [4], and safety requires contextual judgement beyond string matching [5].

We post-train Qwen3-1.7B [1] through four domain-specialized models and one group model. Mathematics, knowledge, and multilinguality use SFT followed by GRPO where automatic verification is reliable; safety relies on SFT with diverse safety data. Overall, our findings suggest that compact models can be improved efficiently, but performance depends strongly on reward reliability, dataset alignment and the balance between specialization and generalization.

## 2 Method

All models except safety use a two-stage strategy: supervised fine-tuning (SFT) followed by Group Relative Policy Optimization (GRPO) [2]. SFT teaches the model the required output format and exposes it to domain-specific examples, while GRPO then directly optimizes answer correctness through a verifiable reward signal. Safety is excluded from the GRPO stage because determining whether a response is "safe" requires contextual judgment rather than string matching, making automatic verification unreliable.

All GRPO stages and the multilingual SFT use LoRA [6] for parameter efficiency, while math and knowledge SFT use full fine-tuning. Reward functions are domain-specific: math and knowledge use shaped rewards with bonuses for well-formed outputs and penalties for missing or multiple boxed answers, while the multilingual model uses a simpler verifiable reward ($+1.0$ correct, $-0.5$ incorrect, $-1.0$ no `\boxed{}`).

For math and knowledge, SFT is performed on reasoning traces to encourage step-by-step thinking. Math traces are provided directly by the training datasets, while knowledge traces are distilled from a larger model ( Qwen3-30B-A3B-Thinking-2507-FP8).

**Mathematics.** We restrict GRPO to the GSM8K dataset because its answers are always clean integers, making automatic verification reliable. Competition-style datasets such as MATH contain answers in symbolic form (e.g. $\frac{\sqrt{3}}{2}$) where string equality is insufficient and normalization would introduce noise into the reward signal.

**Knowledge.** Additionally, few-shot chain-of-thought prompts are used at inference time, selected from validation ablations, to improve format consistency without changing the trained weights.

**Group Model.** The final model is obtained by merging four task-specific experts (multilingual, mathematics, general knowledge, and safety), all derived from Qwen3-1.7B, using the TIES algorithm implemented in MergeKit [7]. TIES combines the task vectors of the individual models while mitigating conflicting parameter updates

through sign-consensus filtering. All experts are merged with equal weights and a density of 0.5. A unified chat template is then applied to ensure consistent prompting and answer formatting across tasks.

## 3 Experimental Setup

All models are run on a single A100 40Gb GPU and logged with Weights & Biases.

### 3.1 Data

All datasets are converted to a unified multiple-choice chat format in which the assistant's final answer is written as `\boxed{X}`, with uppercase labels A–T to match the leaderboard parser. We remove examples overlapping with validation sets, concatenate and shuffle each domain-specific training mixture, and keep restricted validation questions private when required. All data sources are public research benchmarks or derived synthetic examples; nevertheless, because they may contain cultural, educational, linguistic, or annotation biases, we treat the resulting models as benchmark-optimized systems rather than reliable sources of factual or safety-critical truth.

**Mathematics.** The mathematics model is trained on four public reasoning datasets: GSM8K [8], with about 7.5k grade-school arithmetic problems; MATH [9], with about 7.5k competition-level problems across algebra, geometry, number theory, combinatorics, and calculus; NuminaMath-CoT [10], from which we use 50k olympiad-style chain-of-thought examples; and MetaMathQA [11], from which we use 50k augmented GSM8K/MATH rephrasings. After formatting and shuffling, the mixture contains approximately 107k training examples.

**Knowledge.** The knowledge model uses MMLU-Pro/MMLU-style academic questions [3], ScienceQA [12], SciQ [13], and CommonsenseQA [14]. These sources jointly cover broad academic knowledge, science reasoning, and commonsense reasoning. To reduce answer-position bias and better match the leaderboard format, we shuffle answer options and augment mostly four-choice questions by sampling distractors from the same category, randomly varying the number of choices from 2 to 20. Validation uses 1k held-out examples sampled from GPQA Diamond [15] and MMLU-Pro, which resemble the target benchmark; GPQA questions are kept private and are not published in the repository or report.

**Multilingual.** For multilingual training, we combine multiple-choice benchmarks covering Italian, Spanish, Chinese, Russian, and Hindi. INCLUDE [16] provides regional knowledge questions; M3Exam [17], MMMLU [18], Global-MMLU [4], MMLU Russian [19], MILU [20], and C-Eval [21] provide national-exam and academic MCQs. We additionally reserve Kaleidoscope [22] for out-of-distribution evaluation only. To increase task diversity, each four-option question is augmented into three- and two-option variants while preserving the correct answer. SHA-1 deduplication reduces the corpus from 161,904 raw examples to 108,658 unique examples, which are split 90/10 into train and validation sets stratified by language.

**Safety.** The safety model uses three complementary sources. SafetyBench [5], including English and Chinese dev/test splits, covers safety judgements such as illegal activity, physical harm, privacy violations, and ethical misconduct; it is upsampled by a factor of five to increase its weight. PKU-SafeRLHF is reformatted into binary MCQs where the model selects the safer of two responses; labels are derived from per-response safety annotations when they disagree, and from the overall preference otherwise. We sample 30k training examples from PKU-SafeRLHF to cover adversarial and borderline cases. Finally, 2k synthetic examples are generated from 10 hand-written safety prototypes. The three sources are concatenated, shuffled, and split 95/5 into training and evaluation sets.

### 3.2 Evaluation

The baseline for all domains is Qwen3 1.7B base with an additional instruction in the chat template to output in the expected \boxed{X} format. Official leaderboard scores are obtained from the course CI pipeline and an additional benchmark score is obtained from our own domain-specific benchmarked as described below.

For the math model, pass@8 is reported while pass@1 is reported for the other models. The predicted answer is extracted from the first occurrence of the `\boxed{X}` pattern using a regular expression. A prediction is considered correct if the extracted letter matches the ground-truth answer. The same boxed-answer parser as the course CLI is used.

**Mathematics.** We report accuracy on two benchmarks. MATH500 is a 500-problem subset of the

MATH test set spanning all difficulty levels and subject areas, used for local evaluation throughout development. AIME 2024 consists of 30 competition problems and serves as a harder out-of-distribution probe.

**Multilingual.** Evaluation is performed on two benchmarks. The first is `val_vert`, a held-out validation set obtained from the same multilingual data sources used during training. To account for generation stochasticity, inference is repeated with three random seeds (42, 123, and 456) using a sampling temperature of 0.6, and results are reported as mean accuracy $\pm$ standard deviation. Accuracy is reported both globally and separately for each language.

The second benchmark is Kaleidoscope, which is used exclusively for evaluating generalization. Since it originates from a different data distribution and was never used during training, validation, or model selection, it provides a measure of out-of-distribution performance. Following the benchmark setup, we report overall accuracy as well as separate results for languages observed during training (Hindi, Russian, Spanish, and English) and completely unseen languages (c.f Appendix A.1, Figure 2).

**Knowledge.** We evaluate the general-knowledge model on the held-out GPQA Diamond and MMLU-Pro validation set described above. These datasets have been chosen as by inspection of the few validation samples given, the leaderboard is partly made of those datasets. This also ensures we do not train on the leaderboard dataset which would be cheating. We additionally report the validity rate, defined as the fraction of generations containing exactly one parseable boxed option, and the average output length, which helps diagnose whether a method produces overly short answers or unnecessarily long reasoning traces.

**Safety.** Evaluation is performed on two benchmarks. The first is a held-out split of the Safety-Bench and PKU-SafeRLHF mixture (5% of the combined dataset), used for checkpoint selection and monitoring training dynamics. The second is the course leaderboard evaluation, which constitutes the primary out-of-distribution assessment: it covers safety-relevant questions drawn from a different distribution than the training data, providing a measure of how well the acquired safety reasoning generalises beyond the seen datasets.

### 3.3 Implementation Details

All models use the TRL library for both SFT and GRPO and the AdamW optimizer [23] in BF16. Tables. 1, 2 summarizes the implementations details accross the models. Both Math and Knowledge model uses cosine scheduling with linear warmup of ratio 0.03.

| Hyperparameter | Math | Knowledge | Multilingual | Safety |
|---|---|---|---|---|
| Type | Full | Full | LoRA | LoRA |
| Epochs | 3 | 4 | 2 | 3 |
| Learning rate | 2e-5 | 1e-6 | 2e-5 | 1e-6 |
| Per-device batch | 2 | 4 | 8 | 4 |
| Gradient accumulation | 8 | 4 | 4 | 4 |
| Max seq length | 4096 | 8192 | 2048 | 1024 |
| Training examples | 107k | 10k | 288k | 32k |
| Training time | ∼8 hrs | ∼3 hrs | ∼14 hrs | ∼3 hrs |

Table 1: SFT hyperparameters.

| Hyperparameter | Math | Knowledge | Multilingual |
|---|---|---|---|
| Type | LoRA | LoRA | LoRA |
| Generations | 4 | 8 | 4 |
| Epochs | 3 | 4 | 2 |
| Learning rate | 2e-5 | 1e-6 | 2e-5 |
| Per-device batch | 1 | 1 | 4 |
| Gradient accumulation | 8 | 16 | 4 |
| Max seq length | 512 | 1024 | 2048 |
| Training examples | 2.5k | 2k | 20k |
| Training time | ∼8 hrs | ∼5 hrs | ∼13 hrs |

Table 2: GRPO hyperparameters.

**Knowledge.** The temperature is increased from 0.6 to 0.9 for the GRPO. Early stopping on a 10% hold-out split of the training data is implemented for the SFT. The GRPO LoRA modules are q,k,v,o,gate,up,down.

**Multilingual.** The target modules for LoRA for the SFT and GRPO are q,k,v,o,gate,up,down and q,k,v,o respectively. The temperature is increased from 0.6 to 1.2 for the GRPO.

## 4 Results

### 4.1 Main Results

Table 3 reports the group model performance across all four benchmarks.

| Model | Math | Knowledge | Multilingual | Safety |
|---|---|---|---|---|
| **Leaderboard** | | | | |
| Individual models* | 36% | 43% $^{N}$ | 61% | 75% |
| Group model* | 41% | 38% | 61% | 84 % |
| Baseline | 41% | 36% | 59 % | 80 % |
| **Validation set** | | | | |
| Individual models* | 36% | 53.8% | 65.9% | 79.61% |
| Baseline | 31.2% | 52.7% | 41.6% | 76.65 % |

Table 3: Group model results across all four benchmarks. *Ours. $^{N}$ see A.4

**Multilingual.** The base Qwen3-1.7B model achieves $41.6\% \pm 0.2\%$ Pass@1 on the validation set. After supervised fine-tuning, performance in-

creases to $63.1\% \pm 0.4\%$ Pass@1, highlighting the benefit of task-specific multilingual adaptation. Applying GRPO on top of the SFT checkpoint further improves performance to $65.9\% \pm 0.2\%$ Pass@1. The gains, shown in Table 3, are consistent across all target languages. Compared to the pretrained model, SFT substantially improves multilingual multiple-choice performance, while GRPO provides an additional gain across nearly all languages (c.f. Appendix A.1, Figure 1).

The largest improvements are observed for Hindi and Chinese, which are the most challenging languages for the base model. Russian and Spanish achieve the highest final accuracies, exceeding 68% and 71%, respectively. Overall, SFT accounts for most of the performance improvement, while GRPO further refines answer selection and formatting, resulting in the best overall Pass@1 performance.

**Knowledge.** The base Qwen model is a strong baseline, achieving 52.7% pass@1 with high validity. Among all methods, the best result is obtained with LoRA GRPO, which reaches 53.8% pass@1, slightly outperforming the base model (Tab. 7)

Prompting ablations on the base model show that chain-of-thought prompting is competitive with the default baseline. COT1 and COT2 achieve 52.7% and 52.4% pass@1, respectively, while the simpler CoT prompt drops slightly to 50.4%. This suggests that structured reasoning helps preserve performance.

The fine-tuned SFT models perform substantially worse than the base model, despite producing nearly 100% valid outputs. Their very short average generations indicate that they likely learned to produce terse answers without enough reasoning or explanatory context, hurting correctness. However, SFT with reasoning traces improves performance.

GRPO is the strongest training strategy and can surpass the base model. Overall, the main finding is that naive SFT degrades performance, augmented SFT partially recovers it, and carefully tuned GRPO gives the best empirical result.

**Mathematics.** The inclusion of harder datasets such as NuminaMath-CoT and MetaMathQA likely offered limited benefit for a benchmark dominated by simpler arithmetic problems. GRPO produced a small gain when applied to SFT v1, but unexpectedly reduced performance when applied to SFT v2. This may be due to overfitting from the longer SFT stage and a distribution mismatch between the olympiad-style SFT data and the GSM8K-only RL data, causing the model to partially unlearn previously acquired reasoning patterns. In hindsight, a more effective approach would likely have been to first train for several epochs on GSM8K alone, apply GRPO on the same distribution, and only then introduce more challenging datasets if additional performance gains were needed.

**Safety.** SFT improves validation accuracy from 76.65

**Group.** The merged model gives the best overall trade-off: it improves math and safety, keeps multilingual performance stable, but slightly hurts knowledge, indicating both positive transfer and some task interference.

## 5 Ethical Considerations

Adapting the model responsibly would require curated local data, native-speaker evaluation, and contamination checks. Signed-language adaptation would require a separate multimodal system trained and evaluated on signed-language video by fluent signers.

Potential harms include incorrect answers, unsafe recommendations, and biased behavior toward groups poorly represented in the data, especially marginalized communities. We mitigate these risks through domain-specific evaluation, safety-focused training, and by treating benchmark scores as limited evidence rather than proof of real-world reliability. Overall, the project is positive because it studies efficient improvements to small models while explicitly measuring safety and limitations.

## 6 Conclusion

We presented a Qwen3-1.7B post-training pipeline for mathematics, knowledge, multilinguality, and safety. The results show that specialization helps when the training signal matches the evaluation setting: GRPO is most useful for verifiable reasoning, while safety benefits from diverse supervised data. The group model gives the best overall trade-off, especially on safety and multilinguality. The main limitations are GRPO's sensitivity to reward design, distribution mismatch, and the difficulty of exact safety verification. Future work should improve reward design, domain balancing, and evaluation on low-resource languages and broader safety cases.

## References


[1] Qwen Team. Qwen3 technical report, 2025. URL https://arxiv.org/abs/2505.09388.

[2] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z. F. Wu, Zhibin Gou, Zhihong Shao, Zhuoshu Li, Ziyi Gao, Aixin Liu, Bing Xue, Bingxuan Wang, Bochao Wu, Bei Feng, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chong Ruan, Damai Dai, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Hanwei Xu, Honghui Ding, Huazuo Gao, Hui Qu, Hui Li, Jianzhong Guo, Jiashi Li, Jingchang Chen, Jingyang Yuan, Jinhao Tu, Junjie Qiu, Junlong Li, J. L. Cai, Jiaqi Ni, Jian Liang, Jin Chen, Kai Dong, Kai Hu, Kaichao You, Kaige Gao, Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Liang Zhao, Litong Wang, Liyue Zhang, Lei Xu, Leyi Xia, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Mingxu Zhou, Meng Li, Miaojun Wang, Mingming Li, Ning Tian, Panpan Huang, Peng Zhang, Qiancheng Wang, Qinyu Chen, Qiushi Du, Ruiqi Ge, Ruisong Zhang, Ruizhe Pan, Runji Wang, R. J. Chen, R. L. Jin, Ruyi Chen, Shanghao Lu, Shangyan Zhou, Shanhuang Chen, Shengfeng Ye, Shiyu Wang, Shuiping Yu, Shunfeng Zhou, Shuting Pan, S. S. Li, Shuang Zhou, Shaoqing Wu, Tao Yun, Tian Pei, Tianyu Sun, T. Wang, Wangding Zeng, Wen Liu, Wenfeng Liang, Wenjun Gao, Wenqin Yu, Wentao Zhang, W. L. Xiao, Wei An, Xiaodong Liu, Xiaohan Wang, Xiaokang Chen, Xiaotao Nie, Xin Cheng, Xin Liu, Xin Xie, Xingchao Liu, Xinyu Yang, Xinyuan Li, Xuecheng Su, Xuheng Lin, X. Q. Li, Xiangyue Jin, Xiaojin Shen, Xiaosha Chen, Xiaowen Sun, Xiaoxiang Wang, Xinnan Song, Xinyi Zhou, Xianzu Wang, Xinxia Shan, Y. K. Li, Y. Q. Wang, Y. X. Wei, Yang Zhang, Yanhong Xu, Yao Li, Yao Zhao, Yaofeng Sun, Yaohui Wang, Yi Yu, Yichao Zhang, Yifan Shi, Yiliang Xiong, Ying He, Yishi Piao, Yisong Wang, Yixuan Tan, Yiyang Ma, Yiyuan Liu, Yongqiang Guo, Yuan Ou, Yuduan Wang, Yue Gong, Yuheng Zou, Yujia He, Yunfan Xiong, Yuxiang Luo, Yuxiang You, Yuxuan Liu, Yuyang Zhou, Y. X. Zhu, Yanping Huang, Yaohui Li, Yi Zheng, Yuchen Zhu, Yunxian Ma, Ying Tang, Yukun Zha, Yuting Yan, Z. Z. Ren, Zehui Ren, Zhangli Sha, Zhe Fu, Zhean Xu, Zhenda Xie, Zhengyan Zhang, Zhewen Hao, Zhicheng Ma, Zhigang Yan, Zhiyu Wu, Zihui Gu, Zijia Zhu, Zijun Liu, Zilin Li, Ziwei Xie, Ziyang Song, Zizheng Pan, Zhen Huang, Zhipeng Xu, Zhongyu Zhang, and Zhen Zhang. Deepseek-r1 incentivizes reasoning in llms through reinforcement learning. *Nature*, 645(8081):633–638, 2025. ISSN 1476-4687. doi: 10.1038/s41586-025-09422-z. URL http://dx.doi.org/10.1038/s41586-025-09422-z.

[3] Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyan Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhu Chen. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark, 2024. URL https://arxiv.org/abs/2406.01574.

[4] Shivalika Singh, Angelika Romanou, Clémentine Fourrier, David I. Adelani, Jian Gang Ngui, Daniel Vila-Suero, Peerat Limkonchotiwat, Kelly Marchisio, Wei Qi Leong, Yosephine Susanto, Raymond Ng, Shayne Longpre, Wei-Yin Ko, Sebastian Ruder, Madeline Smith, Antoine Bosselut, Alice Oh, Andre F. T. Martins, Leshem Choshen, Daphne Ippolito, Enzo Ferrante, Marzieh Fadaee, Beyza Ermis, and Sara Hooker. Global mmlu: Understanding and addressing cultural and linguistic biases in multilingual evaluation, 2025. URL https://arxiv.org/abs/2412.03304.

[5] Zhexin Zhang, Leqi Lei, Lindong Wu, Rui Sun, Yongkang Huang, Chong Long, Xiao Liu, Xuanyu Lei, Jie Tang, and Minlie Huang. Safetybench: Evaluating the safety of large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15537–15553, 2024.

[6] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021. URL https://arxiv.org/abs/2106.09685.

[7] Prateek Yadav, Derek Tam, Leshem Choshen, Colin Raffel, and Mohit Bansal. Ties-merging: Resolving interference when merging models, 2023. URL https://arxiv.org/abs/2306.01708.

[8] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

[9] Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the math dataset. *arXiv preprint arXiv:2103.03874*, 2021.

[10] Jia Li, Edward Beeching, Lewis Tunstall, Ben Lipkin, Roman Soletskyi, Shengyi Costa Huang, Kashif Rasul, Longhui Yu, Albert Jiang, Ziju Shen, Zihan Qin, Bin Dong, Li Zhou, Yann Fleureau, Guillaume Lample, and Stanislas Polu. Numinamath: Large-scale chain-of-thought math dataset. https://huggingface.co/datasets/AI-MO/NuminaMath-CoT, 2024. Hugging Face dataset.

[11] Longhui Yu, Weisen Jiang, Han Shi, Jincheng Yu, Zhengying Liu, Yu Zhang, James T Kwok, Zhenguo Li, Adrian Weller, and Weiyang Liu. Metamath: Bootstrap your own mathematical questions for large language models. *arXiv preprint arXiv:2309.12284*, 2023.

[12] Pan Lu, Swaroop Mishra, Tony Xia, Liang Qiu, Kai-Wei Chang, Song-Chun Zhu, Oyvind Tafjord, Peter Clark, and Ashwin Kalyan. Learn to explain: Multimodal reasoning via thought chains for science question answering, 2022. URL https://arxiv.org/abs/2209.09513.

[13] Johannes Welbl, Nelson F. Liu, and Matt Gardner. Crowdsourcing multiple choice science questions, 2017. URL https://arxiv.org/abs/1707.06209.

[14] Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. Commonsenseqa: A question answering challenge targeting commonsense knowledge, 2019. URL https://arxiv.org/abs/1811.00937.

[15] David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. Gpqa: A graduate-level google-proof qa benchmark, 2023. URL https://arxiv.org/abs/2311.12022.

[16] Angelika Romanou, Negar Foroutan, Anna Sotnikova, Zeming Chen, Sree Harsha Nelaturu, Shivalika Singh, Rishabh Maheshwary, Micol Altomare, Mohamed A Haggag, Alfonso Amayuelas, et al. Include: Evaluating multilingual language understanding with regional knowledge. *arXiv preprint arXiv:2411.19799*, 2024.

[17] Wenxuan Zhang, Sharifah Mahani Aljunied, Chang Gao, Yew Ken Chia, and Lidong Bing. M3exam: A multilingual, multimodal, multilevel benchmark for examining large language models, 2023. URL https://arxiv.org/abs/2306.05179.

[18] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding, 2021. URL https://arxiv.org/abs/2009.03300.

[19] DragonLLM. MMLU_ru: Russian translation of the mmlu benchmark. https://huggingface.co/datasets/DragonLLM/MMLU_ru, 2023. Hugging Face dataset repository.

[20] Sshubam Verma, Mohammed Safi Ur Rahman Khan, Vishwajeet Kumar, Rudra Murthy, and Jaydeep Sen. MILU: A multi-task Indic language understanding benchmark. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 10076–10132, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-189-6. doi: 10.18653/v1/2025.naacl-long.507. URL https://aclanthology.org/2025.naacl-long.507/.

[21] Yuzhen Huang, Yuzhuo Bai, Zhihao Zhu, Junlei Zhang, Jinghan Zhang, Tangjun Su, Junteng Liu, Chuancheng Lv, Yikai Zhang, Jiayi Lei, Yao Fu, Maosong Sun, and Junxian He. C-eval: A multi-level multi-discipline chinese evaluation suite for foundation models, 2023. URL https://arxiv.org/abs/2305.08322.

[22] Israfel Salazar, Manuel Fernández Burda, Shayekh Bin Islam, Arshia Soltani Moakhar, Shivalika Singh, Fabian Farestam, Angelika Romanou, Danylo Boiko, Dipika Khullar, Mike Zhang, Dominik Krzemiński, Jekaterina Novikova, Luísa Shimabucoro, Joseph Marvin Imperial, Rishabh Maheshwary, Sharad Duwal, Alfonso Amayuelas, Swati Rajwal, Jebish Purbey, Ahmed Ruby, Nicholas Popovič, Marek Suppa, Azmine Toushik Wasi, Ram Mohan Rao Kadiyala, Olga Tsymboi, Maksim Kostritsya, Bardia Soltani Moakhar, Gabriel da Costa Merlin, Otávio Ferracioli Coletti, Maral Jabbari Shiviari, MohammadAmin farahani fard, Silvia Fernandez, María Grandury, Dmitry Abulkhanov, Drishti Sharma, Andre Guarnier De Mitri, Leticia Bossatto Marchezi, Johan Obando-Ceron, Nazar Kohut, Beyza Ermis, Desmond Elliott, Enzo Ferrante, Sara Hooker, and Marzieh Fadaee. Kaleidoscope: In-language exams for massively multilingual vision evaluation, 2025. URL https://arxiv.org/abs/2504.07072.

[23] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization, 2017. URL https://arxiv.org/abs/1412.6980.

# A Appendix

This appendix reports additional experimental details and results that complement the main paper.

## A.1 Multilingual

To further assess multilingual generalization, we evaluate the models on the Kaleidoscope benchmark, which contains examples originating from a different distribution than the training corpus.

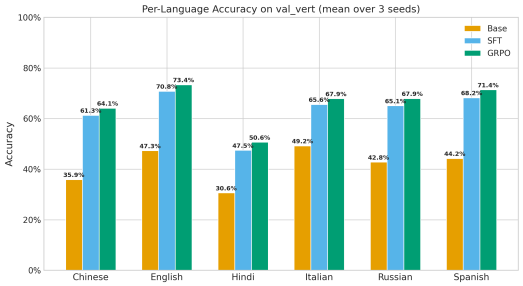



Figure 1: Accuracy on the val vert validation set for each target language. The figure shows the progression from the pretrained model to SFT and GRPO.

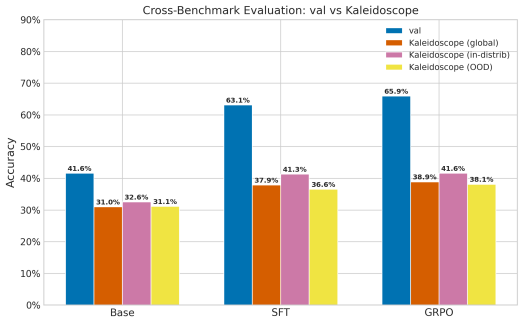



Figure 2: Generalization performance on val vert and Kaleidoscope. In-distribution languages correspond to languages seen during training, while OOD refers to unseen languages.

## A.2 Mathematics

| Hyperparameter | SFT v1 | SFT v2 |
|---|---|---|
| Base model | Qwen3-1.7B | |
| Fine-tuning | Full | |
| Epochs | 1 | 3 |
| Learning rate | 2e-5 | 1e-5 |
| Warmup ratio | 0.05 | 0.03 |
| Effective batch | 16 | |
| Max sequence length | 2048 | 4096 |
| Training examples | ∼107k | |
| Training time | ∼8 h | ∼11 h |

Table 4: Math SFT hyperparameters.

| Hyperparameter | GRPO v1 | GRPO v2 |
|---|---|---|
| Base checkpoint | SFT v1 | SFT v2 |
| Dataset | GSM8K | |
| Training examples | 200 | 2500 |
| Learning rate | 5e-7 | |
| Generations ($G$) | 4 | |
| Max completion length | 512 | |
| Effective batch size | 8 | |
| Precision | BF16 | |
| Training time | 30 min | ∼8 h |

Table 5: Math GRPO hyperparameters.

| Model | MATH500 | AIME24 | Leaderboard |
|---|---|---|---|
| SFT v1 | 28 % | 1.11 % | 32 % |
| SFT v2 | 30 % | 1.11 % | 34 % |
| GRPO v1 | 32 % | 3.33 % | 36 % |
| GRPO v2 | 29 % | 1.11 % | 28 % |

Table 6: Mathematics benchmark results.

## A.3 General Knowledge

| Method | Pass@1 | Valid (%) | Avg. Length |
|---|---|---|---|
| Qwen base | 52.7 | 96.4 | 5042 |
| LoRA SFT old | 39.4 | 100.0 | 5 |
| LoRA SFT | 25.6 | 100.0 | 5 |
| Full SFT | 28.9 | 100.0 | 13 |
| LoRA GRPO | 50.6 | 98.2 | 3671 |
| LoRA SFT augmented | 43.4 | 82.2 | 7969 |
| Full SFT augmented | 48.0 | 82.7 | 8235 |
| Full SFT augmented, small LR | 49.3 | 91.8 | 7276 |
| LoRA GRPO Better | **53.8** | 95.6 | 5010 |

Table 7: General knowledge results on a mix of GQPA and MMLU-PRO (1000 samples).

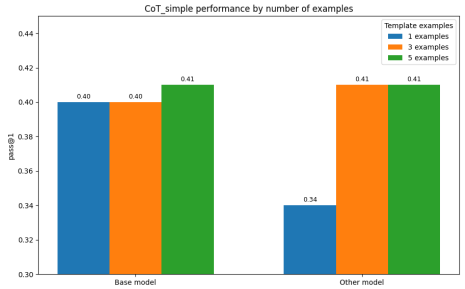



Figure 3: Impact of the number of few-shot examples.

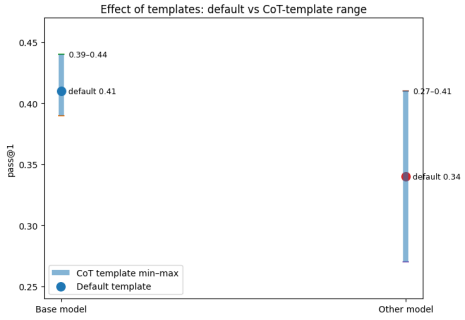



Figure 4: Comparison of chain-of-thought templates.

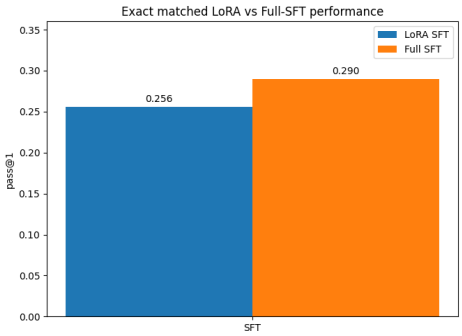



Figure 5: Full fine-tuning versus LoRA adaptation.

### A.4 Submission note - General Knowledge

The submission will be run in the last CI on Wednesday as the Tuesday and Wednesday noon CIs did not run. Therefore, the previous best score is shown. However, this score does not appear in the general knowledge CI's because I pushed my SFT on the group model by mistake on June 02 20:58:08. So, the score shown is the score of the general knowledge benchmark of this group model because the benchmarks are the same. These weights are old and were trained on different datasets but no CI runs have run the actual submission which might cause a mismatch. The rest of the report including the internal validations are run with the proper model.